MODELSENTINEL

Production Drift Monitoring, Root-Cause Analysis, and Retraining Workflow for Deployed Credit Risk ML

Detect drift. Measure impact. Trigger recovery. Re-deploy with confidence.

THE PROBLEM

- Models degrade after deployment
- Production data no longer matches training data
- Silent drift reduces prediction reliability
- Teams need visibility into what changed, why it matters, and how to recover

- Hard to detect distribution shift
- Hard to trust stale predictions
- Hard to know when retraining is necessary

THE SOLUTION

ModelSentinel: Full ML Monitoring Lifecycle

An end-to-end MLOps system for monitoring feature drift, quantifying model degradation, analyzing risk, and orchestrating retraining.

1 BASELINE DATA
- Historical credit-risk training dataset
- reference feature distributions
- initial model: CreditRisk_v2.1

2 DEPLOY MODEL
- production scoring API
- live feature ingestion
- deployed credit-risk predictions

3 MONITOR PRODUCTION
- stream synthetic production batches
- compare baseline vs current distributions
- calculate PSI and KS for each feature

4 DETECT DRIFT
- identify drifted features such as:
  - credit_score
  - income
  - debt_ratio
  - loan_amount
- classify as stable / warning / critical

5 ANALYZE IMPACT
- model health score
- baseline accuracy vs current accuracy
- alert log and root-cause insights
- explain how changing applicant populations affect predictions

6 TRIGGER RETRAINING
- load recent production window
- preprocess features
- train logistic regression model
- run 3-fold cross validation
- evaluate validation accuracy and log loss

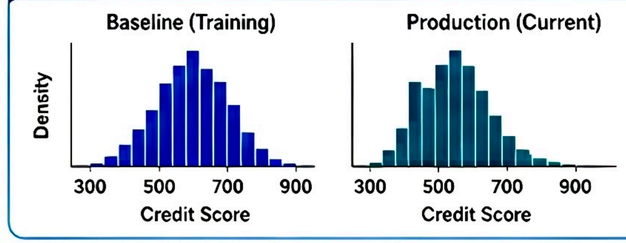
7 REGISTER NEW MODEL
- versioned artifact creation
- model registry update
- deploy CreditRisk_v2.2

8 VERIFY RECOVERY
- drift scores fall
- alerts clear
- health improves
- monitoring continues after redeployment

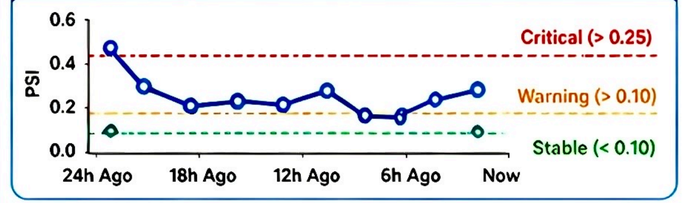
FEATURE DRIFT TABLE

Feature	PSI	KS	Status	Trend (24h)
credit_score	0.39	0.28	CRITICAL	↗
income	0.17	0.14	WARNING	↗
debt_ratio	0.08	0.07	STABLE	→
loan_amount	0.45	0.31	CRITICAL	↗
employment_length	0.05	0.04	STABLE	→

DISTRIBUTION COMPARISON



PSI TREND MONITORING (LAST 24H)



AI ANALYSIS PANEL

Root Cause Analysis:
Increase in lower credit scores and higher debt ratios in recent applicant population.

Business Impact:
Higher risk approvals may increase default rates and financial loss.

Recommended Actions:
Trigger retraining with recent data and monitor high-risk segments closely.

RETRAINING WORKFLOW

Loading Recent Data → Pre-processing Features → 3-Fold Cross Validation → Evaluate (Accuracy & Log Loss) → Deploy New Model

WHAT YOU CAN VERIFY

DRIFT DETECTION
PSI and KS computed from real production batches.

MODEL HEALTH
Baseline vs current accuracy degradation quantified.

RETRAINING AUDIT TRAIL
Versioned artifacts and registry entries for full traceability.

RECOVERY VALIDATION
Post-retraining drift reduction and improved health.

THE TAKEAWAY

ModelSentinel transforms deployed ML from a black box into a monitored, measurable, and recoverable production system.

- Monitor continuously
- Detect drift early
- Retrain with evidence

React + TypeScript + Vite | FastAPI | PSI / KS Drift Metrics | Synthetic Production Simulator | Logistic Regression Retraining | Model Registry

Graduate Capstone / Portfolio Project
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